625 Sampling Project

Ellen Hickman, Mi Huynh, Maze Nead & Zupeng Zeng

2026-01-21

Download data

```
schoolframe <- read.csv("https://raw.githubusercontent.com/ellenthickman/625sampling/refs/heads/main/MI_
```

Week 3: Sampling Frame Analysis

1. Noncoverage

Examine by: searching current # of MI schools (eg. charter). See: https://www.mischooldata .org/student-enrollment-counts-report/

Prevalence:

Solution: Acknowledge and accept in analysis.

2. Blanks: Ellen.

Examine by: Missingness check.

```
# a. Check missingness across variables as blank ""
sapply(schoolframe, function(x) sum(x == "", na.rm = TRUE))
```

```
##            BCODE            BNAME            DCODE    District_Name
##                0                0                0              559
##           Region        County_ID      County_Name Public.nonpublic
##                0                0                0                0
##          g7_totl          g8_totl          g9_totl         g10_totl
##                0                0                0                0
##         g11_totl         g12_totl          tot_all
##                0                0                0
```

Out of 2,443 schools in Michigan, 559 do not have listed District Names. Let's examine this prevalence by Non-Public/Public schools and by Region.

Prevalence: (eg. Frequency tables)

```
# b. Since missing district name might vary by public/private school status, let's check:
tab1 <- table(schoolframe$District_Name == "", schoolframe$Public.nonpublic); kable(tab1)
```

	nonpublic	Public
FALSE	0	1884
TRUE	547	12

```r
# c. By region, let's check:
tab2 <- table(schoolframe$District_Name == "", schoolframe$Region); kable(tab2)
```

	1	2	3	4	5	6	7	8	9
FALSE	19	26	28	26	63	108	479	425	710
TRUE	1	4	5	5	17	25	165	124	213

```r
tab3 <- table(factor(schoolframe$District_Name == "", levels = c(TRUE, FALSE)),
              factor(schoolframe$Region))
kable(prop.table(tab3, margin = 2))
```

	1	2	3	4	5	6	7	8	9
TRUE	0.05	0.1333333	0.1515152	0.1612903	0.2125	0.1879699	0.2562112	0.2258652	0.2307692
FALSE	0.95	0.8666667	0.8484848	0.8387097	0.7875	0.8120301	0.7437888	0.7741348	0.7692308

```r
# c. By district code, let's check:
tab4 <- table(schoolframe$DCODE, schoolframe$District_Name == "")
```

From Tables 1-3, we see that the majority of missing district name cases (547/559, 98%) belong to non-public (ie. private) schools, whereas only a handful of public schools are missing district names (12/1884, ~0.0%). It also looks like even though the private schools are missing District Names, they do belong to a District Code that other, public schools are also a part of.

Solution:

So, the solution to the problem is to 1) ignore District Names as a cluster variable, and rely on region or district code if needed. If need be, the district name could be 2) imputed from the dcode variable for reference in communicating the results.

3. Duplicates: Mi. In the whole observation.

Examine by: "print unique"

Prevalence:

Solution:

4. Clustering: Zupeng.

Examine by:

Prevalence:

Solution:

```r
# Basic cleaning: ensure head-count columns are numeric (handles commas if present)
count_vars <- c("g7_totl","g8_totl","g9_totl","g10_totl","g11_totl","g12_totl","tot_all")

schoolframe <- schoolframe %>%
  mutate(across(all_of(count_vars),
                ~ as.numeric(gsub(",", "", as.character(.x)))))

# Sanity checks
N_rows     <- nrow(schoolframe)
```

```r
N_bcode    <- n_distinct(schoolframe$BCODE)
N_district <- n_distinct(schoolframe$DCODE)

N_rows; N_bcode; N_district
```

```
## [1] 2443
```

```
## [1] 2443
```

```
## [1] 773
```

```r
# Check whether BCODE is unique (should be if each row is a unique school/building record)
bcode_dupes <- schoolframe %>%
  count(BCODE) %>%
  filter(n > 1) %>%
  arrange(desc(n))

bcode_dupes
```

```
## [1] BCODE n
## <0 rows> (or 0-length row.names)
```

```r
# District-level school clustering
district_summary <- schoolframe %>%
  group_by(DCODE) %>%
  summarise(
    n_schools = n(),
    district_name_count = n_distinct(na.omit(District_Name)),
    total_enroll = sum(tot_all, na.rm = TRUE),
    .groups = "drop"
  )

# Distribution of schools per district
summary(district_summary$n_schools)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.00    1.00    2.00    3.16    3.00  129.00
```

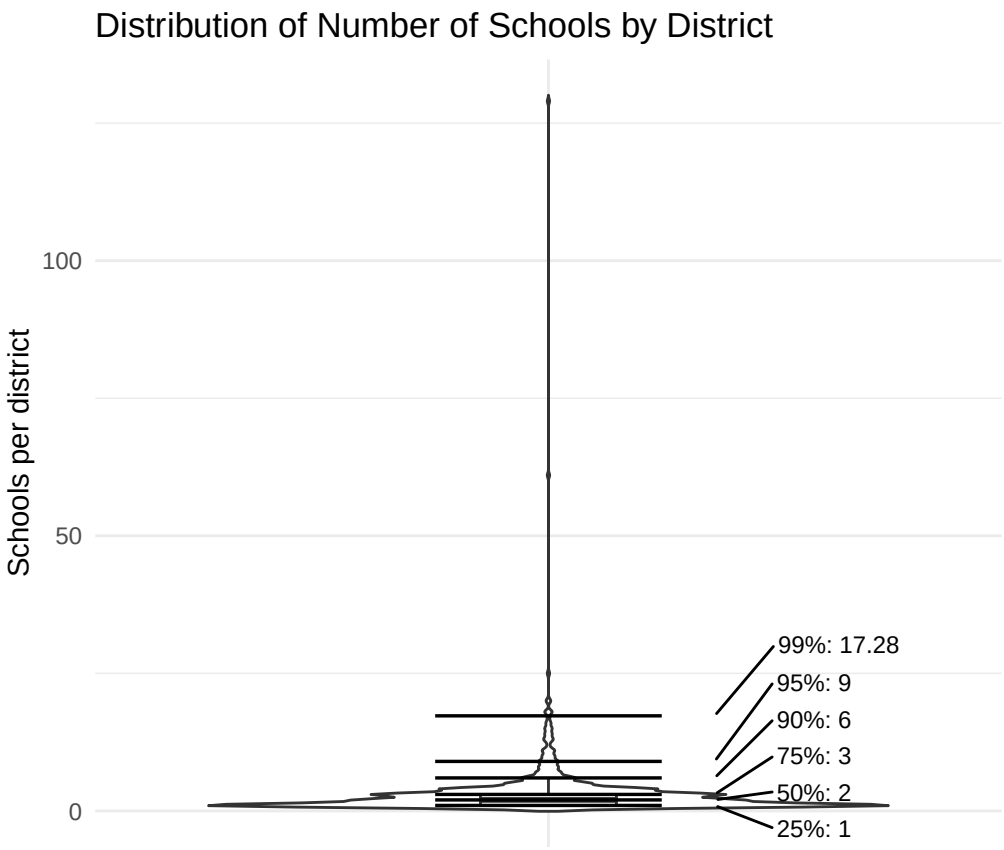
```r
qs <- quantile(
  district_summary$n_schools,
  probs = c(.25, .50, .75, .90, .95, .99),
  na.rm = TRUE
)
qs
```

```
##   25%   50%   75%   90%   95%   99%
##  1.00  2.00  3.00  6.00  9.00 17.28
```

```r
# Violin Plot
q_df <- tibble(
  prob  = names(qs),
  value = as.numeric(qs)
)

ggplot(district_summary, aes(x = "", y = n_schools)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.18, outlier.shape = NA) +
```

```r
geom_segment(
  data = q_df,
  aes(x = 0.85, xend = 1.15, y = value, yend = value),
  inherit.aes = FALSE,
  linewidth = 0.6
) +
geom_text_repel(
  data = q_df,
  aes(x = 1.22, y = value, label = paste0(prob, ": ", round(value, 2))),
  inherit.aes = FALSE,
  direction = "y",
  nudge_x = 0.08,
  hjust = 0,
  size = 3,
  min.segment.length = 0
) +
labs(
  title = "Distribution of Number of Schools by District",
  x = NULL,
  y = "Schools per district"
) +
coord_cartesian(clip = "off") +
theme_minimal() +
theme(
  plot.margin = margin(5.5, 90, 5.5, 5.5),
  axis.text.x = element_blank(),
  axis.ticks.x = element_blank()
)
```




```r
# Largest districts by number of schools
district_summary %>%
  arrange(desc(n_schools)) %>%
  slice_head(n = 10)
```

```
## # A tibble: 10 x 4
##    DCODE n_schools district_name_count total_enroll
##    <int>     <int>               <int>        <dbl>
##  1 82010       129                   3        44562
##  2 41010        61                   2        11888
##  3 81010        25                   2         9391
##  4 50210        20                   2        15535
##  5 82030        20                   2         9848
##  6  9010        18                   2         4847
##  7 39010        18                   2         5929
##  8 82160        18                   2         6790
##  9 73010        17                   2         4669
## 10 33020        16                   2         6810
```

```r
# Potential data issue: DCODE mapping to more than one District_Name
district_summary %>%
  filter(district_name_count > 1) %>%
  arrange(desc(n_schools))
```

```
## # A tibble: 244 x 4
##    DCODE n_schools district_name_count total_enroll
##    <int>     <int>               <int>        <dbl>
##  1 82010       129                   3        44562
```

```
##  2 41010        61                   2        11888
##  3 81010        25                   2         9391
##  4 50210        20                   2        15535
##  5 82030        20                   2         9848
##  6  9010        18                   2         4847
##  7 39010        18                   2         5929
##  8 82160        18                   2         6790
##  9 73010        17                   2         4669
## 10 33020        16                   2         6810
## # i 234 more rows
```

```r
# Duplicates of school name within a district
name_dupes_within_district <- schoolframe %>%
  filter(!is.na(District_Name), !is.na(BNAME)) %>%
  group_by(DCODE, District_Name, BNAME) %>%
  summarise(n_records = n(), .groups = "drop") %>%
  filter(n_records > 1) %>%
  arrange(desc(n_records))

name_dupes_within_district
```

```
## # A tibble: 1 x 4
##   DCODE District_Name           BNAME                   n_records
##   <int> <chr>                   <chr>                       <int>
## 1 82957 Hope of Detroit Academy Hope of Detroit Academy         2
```

5. Many-to-many matching: Maze.

Examine by:

Prevalence:

Solution:

Week 4: SRS Sample Size

```r
cv_target <- 0.05

p_cig  <- 0.25  # ever smoked one cigarette
p_mj   <- 0.18  # ever smoked marijuana
mu_age <- 14    # age first approached
sd_age <- 10

# Element variances
S2_cig <- p_cig * (1 - p_cig)
S2_mj  <- p_mj  * (1 - p_mj)
S2_age <- sd_age^2

# Desired sampling variances
varstar_cig <- (cv_target * p_cig)^2
varstar_mj  <- (cv_target * p_mj)^2
varstar_age <- (cv_target * mu_age)^2

# Required n under SRS, ignoring FPC
n_cig <- S2_cig / varstar_cig
n_mj  <- S2_mj  / varstar_mj
```

```r
n_age <- S2_age / varstar_age

# Round up to ensure CV <= target
n_req <- c(
  cig = ceiling(n_cig),
  mj  = ceiling(n_mj),
  age = ceiling(n_age)
)

# Summary
out <- data.frame(
  variable = c("Ever smoked one cigarette (p)", "Ever smoked marijuana (p)", "Age first approached (mean
  expected_estimate = c(p_cig, p_mj, mu_age),
  element_variance_S2 = c(S2_cig, S2_mj, S2_age),
  target_CV = cv_target,
  desired_sampling_variance = c(varstar_cig, varstar_mj, varstar_age),
  required_n_SRS_ignore_FPC = n_req
)

out
```

```
##                          variable expected_estimate element_variance_S2
## cig Ever smoked one cigarette (p)              0.25              0.1875
## mj      Ever smoked marijuana (p)              0.18              0.1476
## age   Age first approached (mean)             14.00            100.0000
##     target_CV desired_sampling_variance required_n_SRS_ignore_FPC
## cig      0.05                0.00015625                      1200
## mj       0.05                0.00008100                      1823
## age      0.05                0.49000000                       205
```

Week 5: Prepare a table of population counts to each of the regions

```r
schoolframe$tot_all <- as.numeric(schoolframe$tot_all)

regcounts <- schoolframe %>%
  as_tibble() %>%
  group_by(Region) %>%
  summarise(Count = sum(tot_all, na.rm = TRUE))
```

Week 6: Interclass Correlation Coefficient (ICC) for clustering

```r
# Calculate Roh from Previous Study
n_prev <- 7500
a_prev <- 150
m_prev <- n_prev / a_prev

deff <- c(cig = 2.5, mj = 2.0, age = 1.7)

roh <- (deff - 1)/(m_prev - 1)

m_prev
```

```
## [1] 50
```

```r
roh
```

```
##        cig         mj        age
## 0.03061224 0.02040816 0.01428571
```

Week 7: Optimal Sampling Plan

```r
budget <- 500000
cost_per_school <- 3000
cost_per_student <- 50

# Optimal m for each variable
m_opt_raw <- sqrt(cost_per_school * (1 - roh) / (cost_per_student * roh))
m_opt <- ceiling(m_opt_raw)   # round up to ensure CV target

# Corresponding optimal number of clusters a and total n
a_opt <- floor(budget / (cost_per_school + cost_per_student * m_opt))
n_opt <- a_opt * m_opt

opt_plan <- data.frame(
  variable = names(roh),
  roh = as.numeric(roh),
  m_opt_raw = as.numeric(m_opt_raw),
  m_opt = as.integer(m_opt),
  a_opt = as.integer(a_opt),
  n_opt = as.integer(n_opt),
  cost_used = as.numeric(a_opt * (cost_per_school + cost_per_student * m_opt))
)

opt_plan
```

```
##   variable        roh m_opt_raw m_opt a_opt n_opt cost_used
## 1      cig 0.03061224  43.58899    44    96  4224    499200
## 2       mj 0.02040816  53.66563    54    87  4698    495900
## 3      age 0.01428571  64.34283    65    80  5200    500000
```

```r
# Use the largest m_opt across variables to ensure CV target is met for all
m_overall <- max(m_opt)
m_overall
```

```
## [1] 65
```

```r
# Corresponding a and n for overall m
a_overall <- floor(budget / (cost_per_school + cost_per_student * m_overall))
n_overall <- a_overall * m_overall

c(m_overall = m_overall, a_overall = a_overall, n_overall = n_overall,
  cost_used = a_overall * (cost_per_school + cost_per_student * m_overall))
```

```
## m_overall a_overall n_overall cost_used
##        65        80      5200    500000
```

```r
# New expected DEFFs under the chosen overall m
deff_new <- 1 + (m_overall - 1) * roh

# New expected SRS variances, ignoring FPC
```

```r
var_srs_new <- c(
  cig = S2_cig / n_overall,
  mj  = S2_mj  / n_overall,
  age = S2_age / n_overall
)

# New expected sampling variances under cluster design
var_clust_new <- deff_new * var_srs_new

# CVs under new design, standard deviation/mean, expected mean values from week4 input
cv_new <- c(
  cig = sqrt(var_clust_new["cig"]) / p_cig,
  mj  = sqrt(var_clust_new["mj"])  / p_mj,
  age = sqrt(var_clust_new["age"]) / mu_age
)

# Compare to precision targets
varstar <- c(cig = varstar_cig, mj = varstar_mj, age = varstar_age)

precision_check <- data.frame(
  variable = c("cig","mj","age"),
  roh = as.numeric(roh),
  m_used = m_overall,
  a_used = a_overall,
  n_used = n_overall,
  deff_new = as.numeric(deff_new),
  S2 = c(S2_cig, S2_mj, S2_age),
  var_srs_new = as.numeric(var_srs_new),
  var_clust_new = as.numeric(var_clust_new),
  var_target = as.numeric(varstar),
  meets_var_target = as.logical(var_clust_new <= varstar),
  cv_new = as.numeric(cv_new),
  meets_cv_target = as.logical(cv_new <= cv_target)
)

precision_check
```

```
##   variable        roh m_used a_used n_used deff_new       S2   var_srs_new
## 1      cig 0.03061224     65     80   5200 2.959184   0.1875 0.00003605769
## 2       mj 0.02040816     65     80   5200 2.306122   0.1476 0.00002838462
## 3      age 0.01428571     65     80   5200 1.914286 100.0000 0.01923076923
##   var_clust_new var_target meets_var_target     cv_new meets_cv_target
## 1  0.0001067013 0.00015625             TRUE 0.04131854            TRUE
## 2  0.0000654584 0.00008100             TRUE 0.04494798            TRUE
## 3  0.0368131868 0.49000000             TRUE 0.01370483            TRUE
```

```r
# Calculate overall sample fraction
N_pop <- 830138
f_overall <- n_overall / N_pop
f_overall
```

```
## [1] 0.006264019
```

- Since the f is small, <0.05, we can keep excluding the FPC in variance calculations for it would have little impact.

Looping over each variable's optimal allocation

```
# Store results
precision_tables <- list()

for(i in 1:nrow(opt_plan)) {

  # Extract optimal design values for this variable
  m_overall <- opt_plan$m_opt[i]
  a_overall <- opt_plan$a_opt[i]
  n_overall <- opt_plan$n_opt[i]

  # New expected DEFFs under this m
  deff_new <- 1 + (m_overall - 1) * roh

  # New expected SRS variances
  var_srs_new <- c(
    cig = S2_cig / n_overall,
    mj  = S2_mj  / n_overall,
    age = S2_age / n_overall
  )

  # Cluster design variances
  var_clust_new <- deff_new * var_srs_new

  # CVs under new design
  cv_new <- c(
    cig = sqrt(var_clust_new["cig"]) / p_cig,
    mj  = sqrt(var_clust_new["mj"])  / p_mj,
    age = sqrt(var_clust_new["age"]) / mu_age
  )

  # Targets
  varstar <- c(cig = varstar_cig, mj = varstar_mj, age = varstar_age)

  # Precision check table
  precision_tables[[ opt_plan$variable[i] ]] <- data.frame(
    variable = c("cig","mj","age"),
    roh = as.numeric(roh),
    m_used = m_overall,
    a_used = a_overall,
    n_used = n_overall,
    deff_new = as.numeric(deff_new),
    S2 = c(S2_cig, S2_mj, S2_age),
    var_srs_new = as.numeric(var_srs_new),
    var_clust_new = as.numeric(var_clust_new),
    var_target = as.numeric(varstar),
    meets_var_target = as.logical(var_clust_new <= varstar),
    cv_new = as.numeric(cv_new),
    meets_cv_target = as.logical(cv_new <= cv_target)
  )
}

precision_tables$cig <- precision_tables$cig %>%
```

```r
  mutate(cost_used = as.numeric(a_used * (cost_per_school + cost_per_student * m_used)))
precision_tables$mj <- precision_tables$mj %>%
  mutate(cost_used = as.numeric(a_used * (cost_per_school + cost_per_student * m_used)))
precision_tables$age <- precision_tables$age %>%
  mutate(cost_used = as.numeric(a_used * (cost_per_school + cost_per_student * m_used)))

# View tables
precision_tables$cig
```

```
##   variable        roh m_used a_used n_used deff_new       S2   var_srs_new
## 1      cig 0.03061224     44     96   4224 2.316327   0.1875 0.00004438920
## 2       mj 0.02040816     44     96   4224 1.877551   0.1476 0.00003494318
## 3      age 0.01428571     44     96   4224 1.614286 100.0000 0.02367424242
##   var_clust_new var_target meets_var_target     cv_new meets_cv_target
## 1 0.00010281989 0.00015625             TRUE 0.04056006            TRUE
## 2 0.00006560761 0.00008100             TRUE 0.04499918            TRUE
## 3 0.03821699134 0.49000000             TRUE 0.01396369            TRUE
##   cost_used
## 1    499200
## 2    499200
## 3    499200
```

```r
precision_tables$mj
```

```
##   variable        roh m_used a_used n_used deff_new       S2   var_srs_new
## 1      cig 0.03061224     54     87   4698 2.622449   0.1875 0.00003991060
## 2       mj 0.02040816     54     87   4698 2.081633   0.1476 0.00003141762
## 3      age 0.01428571     54     87   4698 1.757143 100.0000 0.02128565347
##   var_clust_new var_target meets_var_target     cv_new meets_cv_target
## 1 0.00010466351 0.00015625             TRUE 0.04092207            TRUE
## 2 0.00006539995 0.00008100             TRUE 0.04492791            TRUE
## 3 0.03740193395 0.49000000             TRUE 0.01381399            TRUE
##   cost_used
## 1    495900
## 2    495900
## 3    495900
```

```r
precision_tables$age
```

```
##   variable        roh m_used a_used n_used deff_new       S2   var_srs_new
## 1      cig 0.03061224     65     80   5200 2.959184   0.1875 0.00003605769
## 2       mj 0.02040816     65     80   5200 2.306122   0.1476 0.00002838462
## 3      age 0.01428571     65     80   5200 1.914286 100.0000 0.01923076923
##   var_clust_new var_target meets_var_target     cv_new meets_cv_target
## 1  0.0001067013 0.00015625             TRUE 0.04131854            TRUE
## 2  0.0000654584 0.00008100             TRUE 0.04494798            TRUE
## 3  0.0368131868 0.49000000             TRUE 0.01370483            TRUE
##   cost_used
## 1    500000
## 2    500000
## 3    500000
```

```r
# a is the number of schools
# m is the number of clusters
# 9 strata
# As clusters increase, variance also increase because homogeneity increases
```

```
# A lower CV is better because there's less variation between samples
```

Week 8

1. First-stage sampling fractions for each stratum, where the total number of schools to sample at the first stage in each stratum is defined by your proportionate allocation of the n_opt clusters. A. Proportionate allocation

$$\frac{N_h}{N} = W_h$$

$$n_h = n * \frac{N_h}{N} = n * W_h$$

B. Input known values

```
n <- 4698 # Total overall n
m <- 54 # No. of schools (clusters)
a <- 87 # No. of students within school (cluster size)
regcounts[1:2]
```

```
## # A tibble: 9 x 2
##   Region  Count
##    <dbl>  <dbl>
## 1      1   3561
## 2      2   5474
## 3      3   8631
## 4      4   4855
## 5      5  18907
## 6      6  33133
## 7      7 191992
## 8      8 188830
## 9      9 374755
```

```
#Calculate stratum weight
regcounts$w_h <- regcounts$Count/sum(regcounts$Count)
```

C. Calculate total number of schools to sample at the first stage in each stratum (m)

```
regcounts$m_h_raw <- regcounts$w_h * m
regcounts$m_h_round <- round(regcounts$m_h_raw); sum(regcounts$m_h_round)
```

```
## [1] 52
```

```
m - sum(regcounts$m_h_round) # Number of schools left to allocate after rounding
```

```
## [1] 2
```

```
# Initialize
regcounts$m_h <- regcounts$m_h_round

#Add 1 to the lowest three strata from the largest three strata to achieve 9 strata
regcounts$m_h[regcounts$m_h == 0] <- regcounts$m_h[regcounts$m_h == 0] + 1

# Subtract 1 from region 9
regcounts$m_h[regcounts$Region == 9] <- regcounts$m_h[regcounts$Region == 9] - 1

# Paired
regcounts$m_h_pair <- c(2,2,2,2,2,2,10,10,22) #Evenly allocated to avoid singleton problem. Noting Stra
```

```r
sum(regcounts$m_h)
```

```
## [1] 54
```

```r
sum(regcounts$m_h_pair)
```

```
## [1] 54
```

```r
# Proportionate allocation of schools across strata
regcounts
```

```
## # A tibble: 9 x 7
##   Region  Count     w_h m_h_raw m_h_round   m_h m_h_pair
##    <dbl>  <dbl>   <dbl>   <dbl>     <dbl> <dbl>    <dbl>
## 1      1   3561 0.00429   0.232         0     1        2
## 2      2   5474 0.00659   0.356         0     1        2
## 3      3   8631 0.0104    0.561         1     1        2
## 4      4   4855 0.00585   0.316         0     1        2
## 5      5  18907 0.0228    1.23          1     1        2
## 6      6  33133 0.0399    2.16          2     2        2
## 7      7 191992 0.231    12.5          12    12       10
## 8      8 188830 0.227    12.3          12    12       10
## 9      9 374755 0.451    24.4          24    23       22
```

D. Calculate total number of schools to sample at the first stage in each stratum (n) (EXTRA)

```r
#Calculate sample size within each stratum
regcounts$n_raw <- (n*regcounts$w_h); regcounts
```

```
## # A tibble: 9 x 8
##   Region  Count     w_h m_h_raw m_h_round   m_h m_h_pair  n_raw
##    <dbl>  <dbl>   <dbl>   <dbl>     <dbl> <dbl>    <dbl>  <dbl>
## 1      1   3561 0.00429   0.232         0     1        2   20.2
## 2      2   5474 0.00659   0.356         0     1        2   31.0
## 3      3   8631 0.0104    0.561         1     1        2   48.8
## 4      4   4855 0.00585   0.316         0     1        2   27.5
## 5      5  18907 0.0228    1.23          1     1        2  107.
## 6      6  33133 0.0399    2.16          2     2        2  188.
## 7      7 191992 0.231    12.5          12    12       10 1087.
## 8      8 188830 0.227    12.3          12    12       10 1069.
## 9      9 374755 0.451    24.4          24    23       22 2121.
```

```r
#Round down strata #6 since f_i is closest to .50 threshold & Round to nearest integer for all else
regcounts <- regcounts %>%
  mutate(
    n_raw = (n * Count) / sum(Count),
    n_h = ifelse(Region == 6, floor(n_raw), round(n_raw))
  )

sum(regcounts$n_h); regcounts
```

```
## [1] 4698
```

```
## # A tibble: 9 x 9
##   Region  Count     w_h m_h_raw m_h_round   m_h m_h_pair  n_raw   n_h
##    <dbl>  <dbl>   <dbl>   <dbl>     <dbl> <dbl>    <dbl>  <dbl> <dbl>
## 1      1   3561 0.00429   0.232         0     1        2   20.2    20
```

```
## 2       2   5474 0.00659   0.356          0     1         2   31.0    31
## 3       3   8631 0.0104    0.561          1     1         2   48.8    49
## 4       4   4855 0.00585   0.316          0     1         2   27.5    27
## 5       5  18907 0.0228    1.23           1     1         2  107.    107
## 6       6  33133 0.0399    2.16           2     2         2  188.    187
## 7       7 191992 0.231    12.5           12    12        10 1087.   1087
## 8       8 188830 0.227    12.3           12    12        10 1069.   1069
## 9       9 374755 0.451    24.4           24    23        22 2121.   2121
```

2. Sort the list to achieve implicit stratification within the regions, using tot_all from previous year as MOS A. Sort the list

```
schoolframe$tot_all <- as.numeric(gsub(",", "", as.character(schoolframe$tot_all)))

schoolframe_sorted <- schoolframe %>%
  arrange(Region, tot_all)
```

3. Compute your zone size (Fractional intervals) for systematic PPeS sampling within each of the 9 strata
   A. Zone size

$$zone\ size = \frac{\sum_{i\epsilon h} MOS_{hi}}{n_h}$$

   B. Calculate

```
regcounts$zone_size <- sum(regcounts$Count) / regcounts$n_h; regcounts
```

```
## # A tibble: 9 x 10
##   Region  Count     w_h m_h_raw m_h_round   m_h m_h_pair  n_raw   n_h zone_size
##    <dbl>  <dbl>   <dbl>   <dbl>     <dbl> <dbl>    <dbl>  <dbl> <dbl>     <dbl>
## 1      1   3561 0.00429   0.232         0     1        2   20.2    20    41507.
## 2      2   5474 0.00659   0.356         0     1        2   31.0    31    26779.
## 3      3   8631 0.0104    0.561         1     1        2   48.8    49    16942.
## 4      4   4855 0.00585   0.316         0     1        2   27.5    27    30746.
## 5      5  18907 0.0228    1.23          1     1        2  107.    107     7758.
## 6      6  33133 0.0399    2.16          2     2        2  188.    187     4439.
## 7      7 191992 0.231    12.5          12    12       10 1087.   1087      764.
## 8      8 188830 0.227    12.3          12    12       10 1069.   1069      777.
## 9      9 374755 0.451    24.4          24    23       22 2121.   2121      391.
```

4. Select the allocated number of schools within each stratum using PPeS sampling using systematic selection based on the fractional intervals (Zone size)

```
# Step 5: PPeS systematic selection within each stratum
library(sampling)
selected_schools <- list()

for(r in unique(schoolframe_sorted$Region)) {

  frame_r <- schoolframe_sorted %>%
    filter(Region == r)

  m_h_r <- regcounts$m_h_pair[regcounts$Region == r]
  M_h_r <- sum(frame_r$tot_all, na.rm = TRUE)

  # Skip if no schools allocated to this region
  if(length(m_h_r) == 0 || m_h_r == 0) next

  # PPeS inclusion probabilities within stratum
```

```r
  pik <- m_h_r * frame_r$tot_all / M_h_r

  # to be safe: cap at 1
  pik[pik > 1] <- 1

  # systematic PPeS selection
  s <- UPsystematic(pik = pik)

  # if nothing selected, skip safely
  if(sum(s) == 0) next

  ss_r <- getdata(frame_r, s)
  ss_r$pik <- pik[s == 1]
  ss_r$selected_region <- r

  selected_schools[[as.character(r)]] <- ss_r
}

selected_schools_df <- bind_rows(selected_schools)

selected_schools_df
```

```
##    ID_unit BCODE                                      BNAME DCODE
## 1        5  9566                        CHS-Horizons School 31030
## 2       15  1555                 Hancock Central High School 31010
## 3       23  2040                     Kingsford Middle School 22030
## 4       30  2039                       Kingsford High School 22030
## 5       23  2128                   LANSING CHRISTIAN SCHOOL 33070
## 6       32  1155            Escanaba Area Public High School 21010
## 7        7  2305                      Mackinac Island School 49110
## 8       27  2692                        Newberry High School 48040
## 9       37  8258               Houghton Lake Adult Education 72020
## 10      71  4133                      Tawas Area High School 35030
## 11     107  1018                     East Jordan High School 15060
## 12     133   554                  Cass Technical High School 82010
## 13     312  4733                        Coloma Middle School 11330
## 14     400  1748                 Homer Community High School 13080
## 15     458  1352                Galesburg-Augusta High School 39050
## 16     502  2613                 Morley Stanwood High School 54040
## 17     540  2005                   Kelloggsville High School 41140
## 18     570   420                    BROTHER RICE HIGH SCHOOL 63010
## 19     595  1462                    Grandville Middle School 41130
## 20     615   475                    Byron Center High School 41040
## 21     631  2587                     Mona Shores High School 61060
## 22     644  4462              West Ottawa High School Campus 70070
## 23      68  7016              Saginaw County Juvenile Center 73000
## 24     277  6302                    Laingsburg Middle School 78040
## 25     349  5994                      Goodrich Middle School 25050
## 26     397  1641                    Pattengill Middle School 33020
## 27     436  2920                        Owosso Middle School 78110
## 28     468  1859                          Ithaca High School 29060
## 29     493  8606                  Hartland M.S. at Ore Creek 47060
## 30     513  5012                 DETROIT CATHOLIC CENTRAL HI 63100
## 31     529  1044                        Eastern High School 33020
```

```
## 32     541  1870                        Jackson High School 38170
## 33     153  8307          St. Clair County Learning Academy 74901
## 34     365  4907                N.W. Wayne Trainable Center 82095
## 35     468    29                Davis Aerospace High School 82010
## 36     536  5031                  Clintondale Middle School 50070
## 37     587  2009                        Kelly Middle School 50020
## 38     629  9419 Life Skills Center of Metropolitan Detroit 82978
## 39     666  5887                  Zemmer Junior High School 44010
## 40     699  4339              James R Geisler Middle School 63290
## 41     729   700                  Clarenceville High School 63090
## 42     756  3505                SOUTH CHRISTIAN HIGH SCHOOL 41160
## 43     779  5957                     Richards Middle School 50100
## 44     799  8376             Career Academy/Adult Education 82070
## 45     817  1242              Fitzgerald Senior High School 50090
## 46     832   785                          Cooley High School 82010
## 47     846  1944                John F. Kennedy High School 82150
## 48     859  3521                     South Lyon High School 63240
## 49     870  5419             Southfield-Lathrup High School 63060
## 50     881  1043                           King High School 82010
## 51     891  6491                       Lakeland High School 63220
## 52     900  3256                          Romeo High School 50190
## 53     909  6276                         Canton High School 82100
## 54     917  1950                     John Glenn High School 82160
##                                District_Name Region County_ID    County_Name
## 1                   Public Schools of Calumet      1        31       Houghton
## 2                      Hancock Public Schools      1        31       Houghton
## 3                   Breitung Township Schools      2        22      Dickinson
## 4                   Breitung Township Schools      2        22      Dickinson
## 5                                                  3        21          Delta
## 6                Escanaba Area Public Schools      3        21          Delta
## 7               Mackinac Island Public Schools      4        49       Mackinac
## 8                    Tahquamenon Area Schools      4        48           Luce
## 9              Houghton Lake Community Schools      5        72      Roscommon
## 10                        Tawas Area Schools      5        35          Iosco
## 11                East Jordan Public Schools      6        15     Charlevoix
## 12           Traverse City Area Public Schools      6        28 Grand Traverse
## 13                   Coloma Community Schools      7        11        Berrien
## 14                    Homer Community Schools      7        13        Calhoun
## 15         Galesburg-Augusta Community Schools      7        39      Kalamazoo
## 16           Morley Stanwood Community Schools      7        54        Mecosta
## 17                Kelloggsville Public Schools      7        41           Kent
## 18                                                 7         3        Allegan
## 19                  Grandville Public Schools      7        41           Kent
## 20                 Byron Center Public Schools      7        41           Kent
## 21          Mona Shores Public School District      7        61       Muskegon
## 22          West Ottawa Public School District      7        70         Ottawa
## 23                               Saginaw ISD      8        73        Saginaw
## 24        Laingsburg Community School District      8        78     Shiawassee
## 25                       Goodrich Area Schools      8        25        Genesee
## 26              Lansing Public School District      8        33         Ingham
## 27                       Owosso Public Schools      8        78     Shiawassee
## 28                       Ithaca Public Schools      8        29        Gratiot
## 29               Hartland Consolidated Schools      8        47     Livingston
## 30                                                 8         9            Bay
```

```
## 31             Lansing Public School District      8        33         Ingham
## 32                     Jackson Public Schools      8        38        Jackson
## 33          St. Clair County Learning Academy      9        74      St. Clair
## 34                     Livonia Public Schools      9        82          Wayne
## 35               Detroit City School District      9        82          Wayne
## 36              Clintondale Community Schools      9        50         Macomb
## 37                East Detroit Public Schools      9        50         Macomb
## 38 Life Skills Center of Metropolitan Detroit      9        82          Wayne
## 39                   Lapeer Community Schools      9        44         Lapeer
## 40           Walled Lake Consolidated Schools      9        63        Oakland
## 41              Clarenceville School District      9        63        Oakland
## 42                                                 9        74      St. Clair
## 43                      Fraser Public Schools      9        50         Macomb
## 44                 Highland Park City Schools      9        82          Wayne
## 45                  Fitzgerald Public Schools      9        50         Macomb
## 46               Detroit City School District      9        82          Wayne
## 47                     Taylor School District      9        82          Wayne
## 48               South Lyon Community Schools      9        63        Oakland
## 49          Southfield Public School District      9        63        Oakland
## 50               Detroit City School District      9        82          Wayne
## 51                      Huron Valley Schools      9        63        Oakland
## 52                    Romeo Community Schools      9        50         Macomb
## 53          Plymouth-Canton Community Schools      9        82          Wayne
## 54   Wayne-Westland Community School District      9        82          Wayne
##    Public.nonpublic g7_totl g8_totl g9_totl g10_totl g11_totl g12_totl tot_all
## 1            Public       0       0       2       10       12       20      44
## 2            Public       0       0      55       62       61       71     249
## 3            Public     111     147       0        0        0        0     258
## 4            Public       0       0     147      170      176      182     675
## 5         nonpublic      44      41      46       53       47       63     294
## 6            Public       0       0     217      231      245      235     928
## 7            Public       7       7       6        5        5       11      41
## 8            Public       0       0      87       82       72       82     323
## 9            Public       0       0      34       26       25       29     114
## 10           Public       0       0     132      137      156      161     586
## 11           Public       0       0     109       80       93       88     370
## 12           Public       0       0     671      506      466      435    2078
## 13           Public     158       0       0        0        0        0     158
## 14           Public       0       0      86       69       57       75     287
## 15           Public       0       0     110      101       81       92     384
## 16           Public       0       0     131      123      123       94     471
## 17           Public       0       0     174      154      127      121     576
## 18        nonpublic       0       0     170      190      173      169     702
## 19           Public     429     439       0        0        0        0     868
## 20           Public       0       0     243      263      273      225    1004
## 21           Public       0       0     334      331      332      329    1326
## 22           Public       0       0     684      606      551      458    2299
## 23           Public       1       3      11        5        1        0      21
## 24           Public      93     104       0        0        0        0     197
## 25           Public     167     167       0        0        0        0     334
## 26           Public     239     209       0        0        0        0     448
## 27           Public     254     278       0        0        0        0     532
## 28           Public     107     111     109      117      111       98     653
## 29           Public     433     453       0        0        0        0     886
```

```
## 30        nonpublic       0       0     296      278      284      194    1052
## 31           Public       0       0     526      374      292      215    1407
## 32           Public       0       0     512      390      405      425    1732
## 33           Public       1       5       8       13        8        1      36
## 34           Public      22      15      25       29       26       17     134
## 35           Public       0       0      69       54       41       39     203
## 36           Public     155     144       0        0        0        0     299
## 37           Public     204     174       0        0        0        0     378
## 38           Public       0       0     184      103      101       47     435
## 39           Public     478       0       0        0        0        0     478
## 40           Public     257     290       0        0        0        0     547
## 41           Public       0       0     164      148      138      155     605
## 42        nonpublic       0       0     171      166      161      183     681
## 43           Public     391     412       0        0        0        0     803
## 44           Public       0       0     373      350       72       97     892
## 45           Public       0       0     279      294      248      224    1045
## 46           Public       0       0     490      345      221      120    1176
## 47           Public       0       0     287      343      328      353    1311
## 48           Public       0       0     313      299      326      451    1389
## 49           Public       0       0     377      396      408      347    1528
## 50           Public       0       0     544      502      315      309    1670
## 51           Public       0      29     458      426      429      391    1733
## 52           Public       0       5     454      510      443      456    1868
## 53           Public       0       0     545      534      468      467    2014
## 54           Public       0       0     680      534      449      456    2119
##            pik selected_region
## 1  0.024712160               1
## 2  0.139848357               1
## 3  0.094263792               2
## 4  0.246620387               2
## 5  0.068126521               3
## 6  0.215038814               3
## 7  0.016889804               4
## 8  0.133058702               4
## 9  0.012059026               5
## 10 0.061987624               5
## 11 0.022334229               6
## 12 0.125433857               6
## 13 0.008229510               7
## 14 0.014948540               7
## 15 0.020000833               7
## 16 0.024532272               7
## 17 0.030001250               7
## 18 0.036564024               7
## 19 0.045210217               7
## 20 0.052293846               7
## 21 0.069065378               7
## 22 0.119744573               7
## 23 0.001112111               8
## 24 0.010432664               8
## 25 0.017687867               8
## 26 0.023725044               8
## 27 0.028173489               8
## 28 0.034581369               8
```

```
## 29 0.046920511                8
## 30 0.055711487                8
## 31 0.074511465                8
## 32 0.091722714                8
## 33 0.002113381                9
## 34 0.007866473                9
## 35 0.011917119                9
## 36 0.017552801                9
## 37 0.022190498                9
## 38 0.025536684                9
## 39 0.028061000                9
## 40 0.032111646                9
## 41 0.035516537                9
## 42 0.039978119                9
## 43 0.047140132                9
## 44 0.052364878                9
## 45 0.061346747                9
## 46 0.069037104                9
## 47 0.076962282                9
## 48 0.081541274                9
## 49 0.089701271                9
## 50 0.098037384                9
## 51 0.101735801                9
## 52 0.109660979                9
## 53 0.118231912                9
## 54 0.124395939                9
```

```r
table(selected_schools_df$Region)
```

```
##
##  1  2  3  4  5  6  7  8  9
##  2  2  2  2  2  2 10 10 22
```

```r
# Check whether the number selected matches allocation!!
selected_counts <- selected_schools_df %>%
  group_by(Region) %>%
  summarise(selected_schools = n(), .groups = "drop")

regcounts %>%
  select(Region, m_h_pair) %>%
  left_join(selected_counts, by = "Region")
```

```
## # A tibble: 9 x 3
##   Region m_h_pair selected_schools
##    <dbl>    <dbl>            <int>
## 1      1        2                2
## 2      2        2                2
## 3      3        2                2
## 4      4        2                2
## 5      5        2                2
## 6      6        2                2
## 7      7       10               10
## 8      8       10               10
## 9      9       22               22
```

5. Report first-stage sampling fraction (f) within each of the nine strata under PPes A. First stage sampling

fraction

$$f_i = \frac{MOS_i}{zone\ size}$$

B. Calculate

```
regcounts$f_i <- regcounts$Count / regcounts$zone_size
```

C. Report: First-stage sampling fraction within each stratum under PPeS

```
# Stratum-level first-stage sampling fraction
regcounts %>%
  select(Region, Count, m_h_pair, zone_size, f_i) # stratum-level first-stage fraction: f_h^ = m_h/m_h
```

```
## # A tibble: 9 x 5
##   Region  Count m_h_pair zone_size      f_i
##    <dbl>  <dbl>    <dbl>     <dbl>    <dbl>
## 1      1   3561        2    41507.   0.0858
## 2      2   5474        2    26779.   0.204
## 3      3   8631        2    16942.   0.509
## 4      4   4855        2    30746.   0.158
## 5      5  18907        2     7758.   2.44
## 6      6  33133        2     4439.   7.46
## 7      7 191992       10      764. 251.
## 8      8 188830       10      777. 243.
## 9      9 374755       22      391. 957.
```

```
# School-level PPeS first-stage inclusion probabilities
selected_schools_df %>%
  select(Region, BCODE, tot_all, pik) # school-level first-stage fraction: pik = m_h * tot_all / M_h
```

```
##    Region BCODE tot_all         pik
## 1       1  9566      44 0.024712160
## 2       1  1555     249 0.139848357
## 3       2  2040     258 0.094263792
## 4       2  2039     675 0.246620387
## 5       3  2128     294 0.068126521
## 6       3  1155     928 0.215038814
## 7       4  2305      41 0.016889804
## 8       4  2692     323 0.133058702
## 9       5  8258     114 0.012059026
## 10      5  4133     586 0.061987624
## 11      6  1018     370 0.022334229
## 12      6   554    2078 0.125433857
## 13      7  4733     158 0.008229510
## 14      7  1748     287 0.014948540
## 15      7  1352     384 0.020000833
## 16      7  2613     471 0.024532272
## 17      7  2005     576 0.030001250
## 18      7   420     702 0.036564024
## 19      7  1462     868 0.045210217
## 20      7   475    1004 0.052293846
## 21      7  2587    1326 0.069065378
## 22      7  4462    2299 0.119744573
## 23      8  7016      21 0.001112111
## 24      8  6302     197 0.010432664
## 25      8  5994     334 0.017687867
## 26      8  1641     448 0.023725044
```

```
## 27       8  2920      532 0.028173489
## 28       8  1859      653 0.034581369
## 29       8  8606      886 0.046920511
## 30       8  5012     1052 0.055711487
## 31       8  1044     1407 0.074511465
## 32       8  1870     1732 0.091722714
## 33       9  8307       36 0.002113381
## 34       9  4907      134 0.007866473
## 35       9    29      203 0.011917119
## 36       9  5031      299 0.017552801
## 37       9  2009      378 0.022190498
## 38       9  9419      435 0.025536684
## 39       9  5887      478 0.028061000
## 40       9  4339      547 0.032111646
## 41       9   700      605 0.035516537
## 42       9  3505      681 0.039978119
## 43       9  5957      803 0.047140132
## 44       9  8376      892 0.052364878
## 45       9  1242     1045 0.061346747
## 46       9   785     1176 0.069037104
## 47       9  1944     1311 0.076962282
## 48       9  3521     1389 0.081541274
## 49       9  5419     1528 0.089701271
## 50       9  1043     1670 0.098037384
## 51       9  6491     1733 0.101735801
## 52       9  3256     1868 0.109660979
## 53       9  6276     2014 0.118231912
## 54       9  1950     2119 0.124395939
```

6. Form pseudo-strata to meet paired selection model to estimate sampling variance

```
selected_schools_df <- selected_schools_df %>%
  arrange(Region, tot_all) %>%   # keep the PPS-selected schools in sorted order within region
  group_by(Region) %>%
  mutate(
    order_in_region = row_number(),
    pseudo_stratum_within_region = ceiling(order_in_region / 2) # create pseudo-strata of size 2 within
  ) %>%
  ungroup()

# Unique pseudo-stratum ID across regions
selected_schools_df <- selected_schools_df %>%
  mutate(
    pseudo_stratum_id = paste0("R", Region, "_P", pseudo_stratum_within_region)
  )
```

Week 10

1. For each of the nine strata, compute the required number of students to subsample from each sampled school based on the stratified PPeS design in order to maintain epsem across all (9) strata.

```
# Step 1: Required number of students per sampled school to maintain epsem

# number of students to sample per school in each stratum to maintain epsem
regcounts$b_h_raw <- regcounts$n_h / regcounts$m_h_pair
```

```r
# Round up to ensure we meet the required number of students per stratum
regcounts$b_h <- ceiling(regcounts$b_h_raw)

regcounts %>%
  select(Region, n_h, m_h_pair, b_h_raw, b_h)
```

```
## # A tibble: 9 x 5
##   Region   n_h m_h_pair b_h_raw   b_h
##    <dbl> <dbl>    <dbl>   <dbl> <dbl>
## 1      1    20        2    10      10
## 2      2    31        2    15.5    16
## 3      3    49        2    24.5    25
## 4      4    27        2    13.5    14
## 5      5   107        2    53.5    54
## 6      6   187        2    93.5    94
## 7      7  1087       10   109.    109
## 8      8  1069       10   107.    107
## 9      9  2121       22    96.4    97
```

2. Do each of the schools that you sampled in a given region have the minimum sufficient size, given the stratum-specific subsample sizes computed in Task #1 (Week 10)? Do subsequent schools on the list have the minimum sufficient size? If not, what will you do?

```r
# Step 2: Check if sampled schools have sufficient size
selected_schools_df <- selected_schools_df %>%
  left_join(regcounts %>% select(Region, b_h), by = "Region") %>%
  mutate(
    # check if the total number of students in the school meets the required number to sample
    sufficient_size = tot_all >= b_h
  )

selected_schools_df %>%
  select(Region, BCODE, tot_all, b_h, sufficient_size)
```

```
## # A tibble: 54 x 5
##    Region BCODE tot_all   b_h sufficient_size
##     <dbl> <int>   <dbl> <dbl> <lgl>
##  1      1  9566      44    10 TRUE
##  2      1  1555     249    10 TRUE
##  3      2  2040     258    16 TRUE
##  4      2  2039     675    16 TRUE
##  5      3  2128     294    25 TRUE
##  6      3  1155     928    25 TRUE
##  7      4  2305      41    14 TRUE
##  8      4  2692     323    14 TRUE
##  9      5  8258     114    54 TRUE
## 10      5  4133     586    54 TRUE
## # i 44 more rows
```

```r
# Identify schools that do not meet the sufficient size requirement
selected_schools_df %>%
  filter(!sufficient_size) %>%
  select(Region, BCODE, tot_all, b_h)
```

```
## # A tibble: 2 x 4
##   Region BCODE tot_all   b_h
```

```
##    <dbl> <int>   <dbl> <dbl>
## 1      8  7016      21   107
## 2      9  8307      36    97
```

3. Begin to describe how you will physically select the subsample of students within a given sampled school (or set of linked schools). What will your second-stage sampling rate be for a given school within a given stratum? How will you acquire the updated rosters from each school? What technique will you use to select the sample at the specified second-stage Rate?

```
# Second stage sampling rate
# f / f_i
regcounts$f_h <- (regcounts$m_h*regcounts$n_h)/regcounts$Count
# Overall f for each stratum (f = m_h*n_h)/MOS_h
regcounts$subsample_rate <- regcounts$f_h/regcounts$f_i
regcounts$m_star <- regcounts$subsample_rate*regcounts$Count
```

4. Write down the overall sampling fraction based on the stratified PPeS design, indicating the overall probability of inclusion for a given student, from a given school (or linked set of schools), in a given stratum. Be careful with notation. Keep in mind that the MOS (Measure of Size) values used for the sampled schools at the first stage and the denominator at the second stage (Did you sample a single school? Or a linked set of schools?) will depend on your response to Task #2 above.

```
overall_f <- sum(regcounts$f_h)
# overall_f <- (54*4698)/830138
```

Week 11

Download data

```
studentlist <- read.csv("https://raw.githubusercontent.com/ellenthickman/625sampling/main/sample_school
head(studentlist)
```

```
##   Grade  Surname     Given.name.s.    Teacher
## 1     7    MAGEE          MONICA L Grady Vest
## 2     7   BOODIN NATALIE ELIZABETH Grady Vest
## 3     7 WORZNIAK            LISA M Grady Vest
## 4     7     BELL      KENNETH MARK Grady Vest
## 5     7   RIDDER         ANNA JANE Grady Vest
## 6     7  SANDERS        HORACE H B Grady Vest
```

1. Find subsampling rate of region 7

$$f_{j|i} = \frac{m_h^*}{MOS_h}$$

$$x_i = f_{j|i} * M_h$$

a. Input known values

```
mstar_7 = 12
MOS_7 = 242

M_7 = 219 #Actual size of school. To find x_7
```

b. Calculate

```
fji_7 <- mstar_7 / MOS_7; fji_7
```

```
## [1] 0.04958678
```

```r
x_7 <- fji_7 * M_7; x_7
```

```
## [1] 10.8595
```

2. Find two seperate probabilities

$$x_7 = x_1(1-p) + x_2(p)$$

a. Fractional interval (inverse of subsampling rate)

$$\frac{1}{f_{j|i}}$$

```r
p <- 0.8595

x_7_compare <- 11*(p)+10*(1-p)
x_7; x_7_compare #Verify
```

```
## [1] 10.8595
```

```
## [1] 10.8595
```

3. Report

```r
1/fji_7; p; 1-p
```

```
## [1] 20.16667
```

```
## [1] 0.8595
```

```
## [1] 0.1405
```

With a fractional interval 20.17, select a sample of 11 students with probability 0.86 or a sample of 10 students with probability 0.14. Selecting a proportionate sample with respect to grade.

3. Select The sample is sorted by grade (7->8) then grouped by teacher (random)

a. Proportion of students within each grade

```r
table(studentlist$Grade)
```

```
##
##   7   8
##  93 126
```

```r
set.seed(123)  # for reproducibility

# --- Step 1: Decide sample size (10 or 11) ---
u <- runif(1)

n_total <- ifelse(u <= 0.86, 11, 10)

# --- Step 2: Proportionate allocation by grade ---
grade_counts <- table(studentlist$Grade)
N <- sum(grade_counts)

# proportional allocation
n_by_grade <- n_total * (grade_counts / N)

# round to integers
n_by_grade <- round(n_by_grade)
```

```r
# adjust to ensure sum = n_total
diff <- n_total - sum(n_by_grade)

# --- Step 3: Systematic sampling within each grade ---
sampled_data <- data.frame()

for(g in names(n_by_grade)){

  cluster <- studentlist[studentlist$Grade == g, ]
  N_h <- nrow(cluster)
  n_h <- n_by_grade[g]

  if(n_h == 0) next

  k <- N_h / n_h  # sampling interval

  start <- runif(1, 1, k)

  indices <- floor(start + (0:(n_h - 1)) * k)
  indices <- pmin(indices, N_h)  # avoid overflow

  sampled <- cluster[indices, ]

  sampled_data <- rbind(sampled_data, sampled)
}

# --- Final sample ---
sampled_data
```

```
##     Grade    Surname        Given.name.s.        Teacher
## 14      7     ABELES                 JODI     Grady Vest
## 33      7  WILDSTEIN                 EVAN      Qixuan Li
## 52      7    STEWART    LA SHONDA ANNETTE     Bill Pesau
## 70      7    LANGDON         JOHN DOUGLAS     Bill Pesau
## 89      7    TRECKER              MOLLY A Andrew Bellman
## 102     8      PERRY            BRIDGET E   Joe Williams
## 123     8     LITTLE MICHAEL FREDERICK,JR   Robert McFay
## 144     8    RAINSON               TARA A   Joseph Miden
## 165     8     RAZNIK           JACQUELINE   Joseph Miden
## 186     8 FEIERSTEIN        MATTHEW DAVID    Jame Vogner
## 207     8     LEVINE               LISA S       Jane Doe
```

RECYCLE BIN

A. Overall sampling rate (Lec. 8 PPeS, p. 58)

$$f = \frac{ma*}{\sum_{i\epsilon h} MOS_{hi}}; where m = \#\ of\ clusters\ and a* = targeted\ sub\ sample\ size$$

B. Calculate overall sampling fraction

```r
#Calculate overall sampling rate
f <- (m*a)/sum(regcounts$Count); f
```

```
## [1] 0.0056593
```

C. Strata sampling rate (Lec. 8 PPeS, p. 58)

$$f = \frac{m_h m_h *}{\sum_{i \epsilon h} MOS_{hi}}; where m = \#\ of\ clusters\ and a* = targeted\ sub\ sample\ size$$

D. Calculate

```
regcounts$f_h <- (regcounts$m_h * regcounts$m_h) / sum(regcounts$Count); regcounts
```

```
## # A tibble: 9 x 16
##   Region  Count     w_h m_h_raw m_h_round   m_h m_h_pair  n_raw   n_h zone_size
##    <dbl>  <dbl>   <dbl>   <dbl>     <dbl> <dbl>    <dbl>  <dbl> <dbl>     <dbl>
## 1      1   3561 0.00429   0.232         0     1        2   20.2    20    41507.
## 2      2   5474 0.00659   0.356         0     1        2   31.0    31    26779.
## 3      3   8631 0.0104    0.561         1     1        2   48.8    49    16942.
## 4      4   4855 0.00585   0.316         0     1        2   27.5    27    30746.
## 5      5  18907 0.0228    1.23          1     1        2  107.    107     7758.
## 6      6  33133 0.0399    2.16          2     2        2  188.    187     4439.
## 7      7 191992 0.231    12.5          12    12       10 1087.   1087      764.
## 8      8 188830 0.227    12.3          12    12       10 1069.   1069      777.
## 9      9 374755 0.451    24.4          24    23       22 2121.   2121      391.
## # i 6 more variables: f_i <dbl>, b_h_raw <dbl>, b_h <dbl>, f_h <dbl>,
## #   subsample_rate <dbl>, m_star <dbl>
```

D. Sampling rate of clusters at the first stage in each stratum (n)

$$n_h = \frac{nMOS_i}{\sum_i MOS_i}$$

D. Sampling rate of clusters (schools) at the first stage in each stratum (m)

$$m_h = \frac{mMOS_h}{\sum_{i \epsilon h} MOS_h}$$

F. f_h

$$f_h = \frac{n_h m_h^*}{\sum_{i \epsilon h} MOS_{hi}}$$